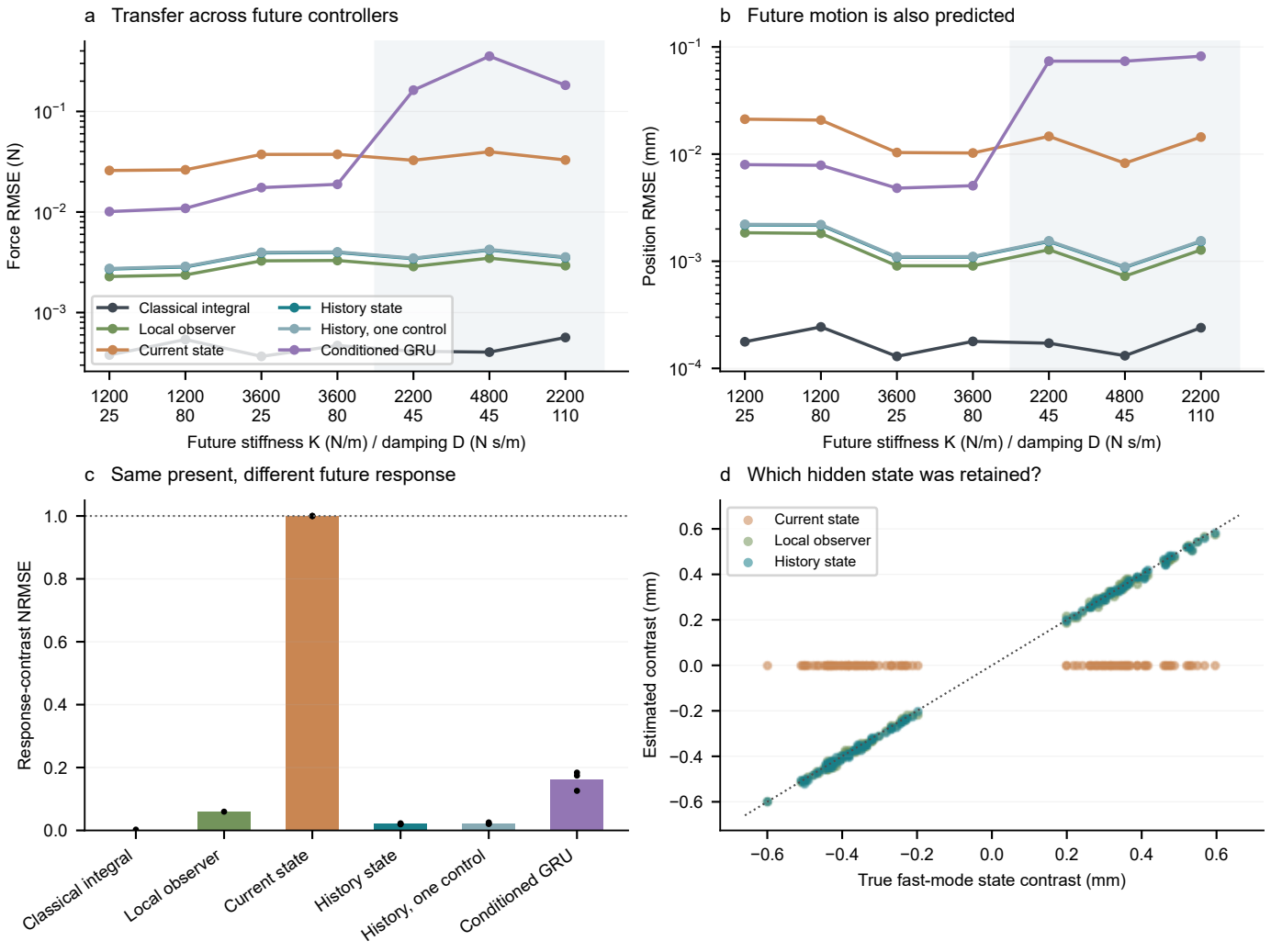

a Transfer across future controllers
Force RMSE (N)
Classical integral
Local observer
Current state
History state
History, one control
Conditioned GRU
1200 25
1200 80
3600 25
3600 80
2200 45
4800 45
2200 110
Future stiffness K (N/m) / damping D (N s/m)
b Future motion is also predicted
Position RMSE (mm)
Future stiffness K (N/m) / damping D (N s/m)
c Same present, different future response
Response-contrast NRMSE
Classical integral
Local observer
Current state
History state
History, one control
Conditioned GRU
d Which hidden state was retained?
Current state
Local observer
History state
Estimated contrast (mm)
True fast-mode state contrast (mm)